Truncated Low-rank and Total p Variation Constrained Color Image Completion and its Moreau Approximation Algorithm

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***Abstract*—Recently, low-rank (LR) and total variation (TV) constrained tensor completion algorithms have been broadly studied for image restoration. These algorithms, however, ignore the difference of the intrinsic properties along spatial structure, spectral correlation, and unfolded mode. In this paper, we go further by providing a detailed comparison of the LR and TV properties in matrix and tensor cases, and figure out the LRTV constraints for pixel matrices are more evident and accordant than for others. This inspires us to develop a simple yet effective multichannel LRTV model that is capable of genuinely discovering the intrinsic properties with reduced computational cost. Moreover, due to the suboptimality of nuclear norm and l_1 norm in approximating the essential low rank and low gradient properties, we employ two enhanced constraints, i.e., truncated nuclear norm (TNN) and total p variation T$_p$V, for a better performance. This results in a challenging problem since that both TNN and T$_p$V are nonsmooth and nonconvex. Observing that the Moreau approximation of T$_p$V constraint is a continuous difference-of-convex function, we then develop a first-order method by repeatedly computing two simple proximal operators. Under mild assumption, we further prove that the sequence generated by our method clusters at a stationary point. Extensive experimental results on color image completion show the efficacy and efficiency of our method over state-of-the-art competitors.**

***Index Terms*—matrix/tensor completion, multichannel image inpainting, low-rank, total variation, Moreau approximation.**

I. INTRODUCTION

IMAGE Completion is a technique of filling in the missing region of incomplete images using the values of reference (available) pixels and the structural assumptions (priors), which is a very active topic in the computer vision community. Research on this topic is boosted by numerous applications such as scene reconstruction [1], video decoding [2], contextual information recognition [3], and recommender systems [4]. Roughly, most existing methods for image completion can be grouped into three categories: deep learning [5][6], nonlocal self-similarity patches [7][8], and prior model [9]. In recent years, the deep learning methods based on convolutional neural networks (CNNs) or generative adversarial networks (GANs) has pushed aside patch- and prior-based approaches due to its strong capability of learning extremely powerful hierarchical nonlinear representation of the data [10]. However, the implementation of deep learning methods requires massive samples to train the net parameters, which is not practical for singleton image restoration tasks. Nonlocal self-similarity patches-based methods assume that a target region can be synthesized by estimating the pixels according to the reconstruction of multiple similar patches. Thus, such kind of approaches is also known as exemplar-based synthesis and broadly used in filling large holes or objective removal [7]. However, the performance of these methods is sensitive to noise and depends heavily on the structural information. When the reference and missing pixels are independent, e.g., white noise, the completion results are generally not desirable [11]. Moreover, these methods are also time-consuming even applied in completing a single image with low resolution.

This paper focuses on the prior-based approaches, which have aroused wide attentions due to its interpretability of intuitive principle and robustness to specific noise. Natural images have many properties, such as sparsity, low-rank (LR), and edge discontinuities, which can be employed as useful priors for designing completion methods. While the sparsity always lies in the transformed domain, i.e., wavelet [12] and curvelet [13], the other two normally lie in the original images, thus are more intuitive and interpretable. Nuclear norm minimization (NNM), the tightest convex relaxation of low-rank, is a powerful tool to capture the global structure information, yet the resulted solution may deviate far from the optimal solution due to its over-penalization of the larger singular values [14]. Bilinear factorization (BF) [15][16], another line for low-rank approximation, suffers from the difficulty of pre-estimating the rank of factor matrices. To avoid such issues, various algorithms [17][18][19][20] attempt to use nonconvex extensions, e.g., Schatten-p norm [17][20], truncated nuclear norm [21][22], and weighted nuclear norm [23], as the low-rank surrogates. All of them have provided a large advantage over nuclear norm, but optimizing the nonconvex problems efficiently is a challenging problem all the time.

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Some recent works [24][25] argue that low-rank constraint cannot fully exploit some underlying local structures, e.g., edge discontinuities or piecewise smoothness, of natural images, and the completion results may be unsatisfactory. As a complementary constraint, the well-known total variation (TV) norm is often incorporated into image recovery for better preservation of piecewise smooth prior [24][25][26]. The rigorous formulation of TV for natural images has long been challenging, but recent advances in nonconvex extensions [27], Shannon interpolation [28], and discretization definition [29] have made the implementation of TV more sound. Yet, the numerical minimization of TV constrained problems remains longing for more attention. Although the development of convex nonsmooth optimization, with efficient primal-dual formulations and alternating directions methods [30][31], have made the TV-based minimization affordable, the usual operations of nested strategy and linear system at each iteration always raise theoretical and practical convergence issues.

Most of the existing completion methods are matrix oriented. However, the natural images are mostly represented as 3-order tensors, where the three dimensions are height, width and color channel (RGB). There are mainly three strategies for color image completion. The first is applying the matrix completion (MC) algorithm to each channel (or termed as pixel matrix) [24][32]. Such a direct scheme will not exploit the spectral correlation among RGB channels, and the results may be unsatisfactory. The second is turning the three channels into a less correlated spectral space, such as YCbCr, then performing completion in each channel of the transformed space [7]. However, the color transform may complicate the missing pixels distribution, and the correlation among RGB channels is not fully exploited. The third, which is the mostly used one, attempts to perform tensor completion (TC) on the color channels jointly for better utility of spectral correlation [11][33][34][35][36]. Two representative works along this line is the high accuracy low-rank (HaLR) TC method [33] and the simultaneously LR and TV constrained method [34], where the former introduces a new definition of the nuclear norm for tensors and the latter applies matrix factorization to the all-mode matricizations of the tensor as an alternative of the tensor nuclear norm.

TC using LRTV constraint is deemed as a more promising way for color image completion [35][36]. The main difference of existing methods lies in the optimization scheme. For example, Refs. [34], [35], and [11] respectively apply block coordinate decent (BCD), alternating direction method of multipliers (ADMM), and primal-dual splitting (PDS) algorithm for solving their proposed problems. In this work, we argue that tensor may be not the best choice for multichannel image completion. As the unfolded matrix at each mode shares the same elements, their nuclear norms are interdependent and the optimal tensor rank is difficult to determine. Although certain auxiliary matrix can be introduced to split the interdependent modes for relief of this issue, the optimality of its solution is still questionable. Moreover, the tensor operations are always time-consuming compared to the matrix manipulations. To remedy these issues, our paper proposes a new joint completion method employing MC instead of TC for color images. Based on the LRTV constraint, we combine total p variation (T_pV) and truncated nuclear norm (TNN) to closely approximate the intrinsic image properties, where T_pV accounts for local piecewise prior and TNN exploits regular global patterns. Numerous experimental results on natural images validate the advantages of our method over state-of-the-art ones. Concretely, the contributions can be summarized as follows:

1) The distributions of singular values and gradients for RGB channels and unfolded matrices are studied by samples. By comparison, we find that the pixel matrices present similar or even clearer image properties than the unfolded tensors. This motivates us to employ MC technique instead of TC for filling natural images, which saves much computational cost while holding similar or even better performance.
2) Since that processing each pixel matrix separately would often results in inferior performance to processing the color channels jointly [36], we further analyze the inter-channel relationships of the LRTV property and discover that they share very close distribution to each other. On that basis, we propose a joint multichannel scheme to perform real color images by grouping the RGB pixel matrices into a unified cost function.
3) Due to the nonconvexity and nonsmoothness of T_pV and TNN constraints, the optimal solution is challenging to solve and standard first-order methods cannot be applied efficiently. We provide a difference-of-convex (DC) based optimizer, together with analysis on its theoretical properties and practical behaviors. Specifically, we approximate the T_pV constraint by its Moreau envelope in each iteration. With simple observation that the resulted function is a continuous DC function, we can then approximately minimize it by suitable majorization techniques.

The rest of this paper is organized as follows. In next section, we review some representative studies for image completion based on LRTV. Subsequently, we present our joint multichannel scheme in Sect. III, and develop an efficient optimization method in Sect. IV. Numerical experimental comparisons with other state-of-the-art image completion algorithms will be reported in Sect. V. Finally, we conclude this paper in Sect. VI. The source code and real images used in this paper are provided as the supplemental material and will be released online after the revision phase.

II. RELATED WORK

Given an incomplete image $\boldsymbol{M} \in \Re^{I_1 \times I_2 \times I_3}$ with Ω indicating the support set of observed indices, the matrix/tensor completion problem under the noise-free environment can be formulated as follows, c.f. [33][34][35]:

$$\min_{\boldsymbol{X}} f(\boldsymbol{X}), \text{ s.t. } P_\Omega(\boldsymbol{X}) = P_\Omega(\boldsymbol{M}), \tag{1}$$

where $\boldsymbol{X} \in \Re^{I_1 \times I_2 \times I_3}$ represents the recovery result, the function $f(\cdot)$ is used to evaluate prior assumption, $P_\Omega(\boldsymbol{X}) = \boldsymbol{Q} \odot \boldsymbol{X}$ denotes the projection operator with $\odot$ being the Hadamard product, and $\boldsymbol{Q} \in \{0,1\}^{I_1 \times I_2 \times I_3}$ is an index tensor

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representing the missing and available entries of $\boldsymbol{M}$ as 0 and 1, respectively.

Note that Eq. (1) has the strict constraint of $P_\Omega(\boldsymbol{X}) = P_\Omega(\boldsymbol{M})$, which means that the observed entries with noise in $P_\Omega(\boldsymbol{M})$ will be intactly copied to $\boldsymbol{X}$, so it is less reasonable for noisy scenario. In this case, the general model under investigation will result in either the regularized least squares problem without constraints [21][37][38],

$$\min_{\boldsymbol{X}} \frac{1}{2}\|P_\Omega(\boldsymbol{X}-\boldsymbol{M})\|^2 + \lambda f(\boldsymbol{X}), \tag{2}$$

where λ is a given parameter; or the constrained problem [11],

$$\min_{\boldsymbol{X}} f(\boldsymbol{X}), \text{ s.t. } \|P_\Omega(\boldsymbol{X}-\boldsymbol{M})\|^2 \leq \delta, \tag{3}$$

where δ is a tolerance parameter that depends on the noise variance of the observed pixels. It is confirmed in [37] that problems (2) and (3) are interchangeable in the sense that a solution of (2) can also be that of (3) under some well-tuned parameters λ, δ, and vice versa. In real applications, the inequality constraint in (3) is relatively more difficult to address than the Lagrange-like problem (2) [11]. Moreover, the noise level of the observed pixels is often unknown. These motivate us to focus on problem (2) in this work. In the sequel, we review prior studies along two directions following (2), i.e., multichannel completion and property approximation.

A. Multichannel Completion

For depth [24] or grey [32] images, i.e., $I_3 = 1$, problem (2) is referred to as matrix completion. For color images with $I_3 = 3$, it is not a trivial extension from single channel to multiple channels. A direct scheme is to conduct MC for each channel independently (MCInd) as

$$\min_{\boldsymbol{X}^{(i)}} \frac{1}{2}\|P_\Omega(\boldsymbol{X}^{(i)}-\boldsymbol{M}^{(i)})\|^2 + \lambda f(\boldsymbol{X}^{(i)}), \tag{4}$$

where $X^{(i)}$, $i = 1,2,3$ represents the ith channel of $\boldsymbol{X}$. This scheme treats the three channels separately and equally in the completion process, thus false colors or artifacts may often be generated [36]. Most recent studies apply tensor technique to fulfill multichannel completion. In [33], a tensor nuclear norm has been discussed to approximate the low-rank property of natural images, in which the function $f(\boldsymbol{X})$ is defined by

$$f(\boldsymbol{X}) = \sum_{i=1}^{3} \beta_i \|\boldsymbol{X}_{(i)}\|_*, \tag{5}$$

where $\beta_i \geq 0$ denotes the weight parameters for individual tensor modes, $\|\cdot\|_*$ is the convex surrogate of rank function, and $\boldsymbol{X}_{(i)} \in \Re^{I_i \times \Pi_{k\neq i} I_k}$ represents the ith mode unfolded matrix of $\boldsymbol{X}$.

Low-rank constraint, albeit useful for discovering global structure, is not applicable to exploit the underlying local properties for completion. The point is particularly obvious for visual data, which often exhibits smooth and piecewise structures in spatial dimension, due to objects or edges therein. [34] considers to introduce the TV regularization into the TC problem (2) and proposes a LRTV constrained model as

$$\min_{\boldsymbol{X}} \frac{1}{2}\|P_\Omega(\boldsymbol{X}-\boldsymbol{M})\|^2 + \sum_{i=1}^{3} \frac{\beta_i}{2}\|\boldsymbol{M}_{(i)} - \boldsymbol{A}_{(i)}\boldsymbol{X}_{(i)}\|_F^2 + \lambda \mathrm{TV}(\boldsymbol{X}_{(3)}), \tag{6}$$

where $\mathrm{TV}(\boldsymbol{X}_{(3)})$ is the isotropic total variation of $\boldsymbol{X}_{(3)}$, and $\boldsymbol{A}_{(i)}\boldsymbol{X}_{(i)}$ is the low-rank matrix factorization to ith mode matricization of $\boldsymbol{X}$. Unfortunately, there exists three drawbacks in Eq.(6): (i) imposing TV on factor matrix $\boldsymbol{X}_{(3)}$ is difficult to comprehend; (ii) the rank of each mode should be specified for factorizations, which is intricate; (iii) the isotropic TV is inflexible to model piecewise smoothness on arbitrary modes. To resolve these issues, [11] and [35] propose two similar LRTV constrained TC problems based on isotropic and anisotropic TV regularizations, respectively:

$$\min_{\boldsymbol{X}} \frac{1}{2}\|P_\Omega(\boldsymbol{X}-\boldsymbol{M})\|^2 + \sum_{i=1}^{3} \frac{\beta_i}{2}\|\boldsymbol{X}_{(i)}\|_* + \sum_{i=1}^{3} \lambda_i \mathrm{TV}(\boldsymbol{X}_{(i)}). \tag{7}$$

Numerical experiments have demonstrated the advantages of model (7) over model (6). In practice, however, both [11] and [35] share two deficiencies. First, the distributions of singular values and total variations change greatly between different modes, which results in exhausting search for best balance parameters λ_i and β_i. Second, suffering from the non-smoothness of both nuclear norm and TV, standard gradient based optimization is not applicable for problem (7). Although the proximal splitting methods are gaining attention, they often converge slowly due to introduction of many auxiliary parameters.

B. Property Approximation

For the low-rank property, most mentioned approaches adopt nuclear norm, i.e., $\|\boldsymbol{Z}\|_* = \sum_i \sigma_i(\boldsymbol{Z})$ with $\sigma_i(\boldsymbol{Z})$ being the ith biggest singular values of matrix $\boldsymbol{Z}$, for a convex relaxation of the original NP-hard problem. Recently, this direction is seeing a paradigm shift, as nonconvex regularizers push aside the convex counterparts. For instance, [39] and [40] respectively utilize the Schatten-p norm and capped nuclear norm to approximate the original LRMC problem. In addition, considering that the singular values should not be penalized equally, Gu et al. [23] develop a weighted nuclear norm to deal with several low-level computer vision problems including image completion. However, all these methods suffer from poor parameter sensitivity. For this reason, Selesnick et al. [41] try to restrict the parameter span for constructing a strictly convex function with nonconvex penalizations, but this trick leads to poorer performance due to its looser approximation of intrinsic properties. More discussions w.r.t various nonconvex relaxation functions can be found in [17], where an iteratively reweighted nuclear norm (IRNN) strategy is developed to solve the general problems with nonconvex regularizers. Yet, IRNN converges slowly due to minimizing a surrogate function based on linearizing loss function and constraints simultaneously [20].

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For the total variation property, two well-known formulations, i.e., isotropic TV (ITV) and anisotropic TV (ATV), are respectively written in Eq.(8) and Eq.(9) in terms of matrix $\mathbf{Z}$.

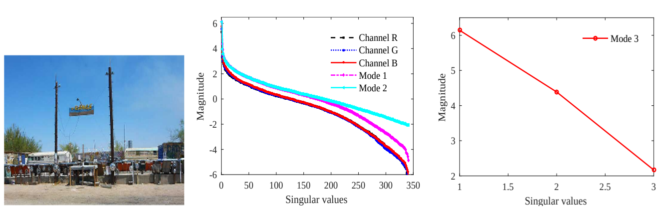
$$\mathrm{ITV}(\mathbf{Z}) = \sum_{i,j} \sqrt{(D_1 Z_{i,j})^2 + (D_2 Z_{i,j})^2}, \tag{8}$$

$$\mathrm{ATV}(\mathbf{Z}) = \sum_{i,j} |D_1 Z_{i,j}| + |D_2 Z_{i,j}|, \tag{9}$$

where $D_n, n = 1, 2$ is an I_n-1-by-I_n matrix with $[D_n]_{i,i} = 1$, $[D_n]_{i,i+1} = -1$ and the other entries are zeros. The studies on TV norm can be roughly broken down into two directions. One line of methods focuses on the rigorous definition for TV norm applying in the discrete field such as images [28], [29], [33], while the other line attempts to discover the intrinsic gradient information [24], [27] using nonconvex surrogates. In this paper, we follow the second line aiming for simultaneously discovering piecewise structures and avoiding complex implementation of rigorous TV definition. In fact, TV norm is a relaxation of the l_0 norm implementing a straightforward measure of the sparsity of gradients. Since the TV norm penalizes large gradient magnitudes, it may over-punish real image edges and boundaries [24]. Therefore, several algorithms directly solving l_0 gradient minimization have been proposed. For example, Ono [42] presents a l_0 gradient projection framework for image smoothing. In [24], the authors argue that though most pixels have zero gradients, there is still a non-ignorable part of pixels whose gradients are small but nonzero. Accordingly, they propose a low (instead of l_0) gradient penalization method and achieve appealing performance on depth images.

III. PROPOSED MODEL

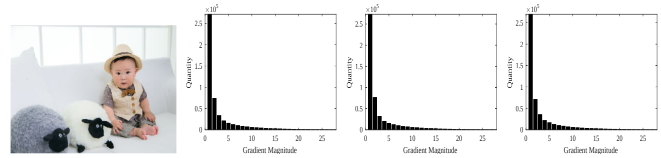
As discussed previously, though joint completion of RGB channels is considered as a promising way for color image inpainting, it is not a trivial extension from grayscale image to multiple channels. Simply concatenating each channel into a long vector [36] or conducting multichannel completion in unfolded modes may all lead to suboptimal results. This section presents a simple yet effective model to overcome the mentioned problems. We start by analyzing the low-rank and low-gradient structure on color images and then describe how this model can be enhanced by certain nonconvex regularizations.



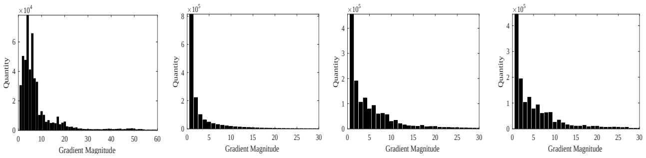
(a) Original image (b) RGB and modes 1, 2 (c) Mode 3

Fig. 1. The singular value distributions of an RGB image in each channel and mode. For clarity, the plots of (b) and (c) are exhibited in log domain.

Fig. 1 illustrates the singular value distributions of a natural image with size 341×600. Note that the number of singular



(a) Original image (b) Channel R (c) Channel G (d) Channel B



(e) Inter-channels (f) Mode 1 (g) Mode 2 (h) Mode 3

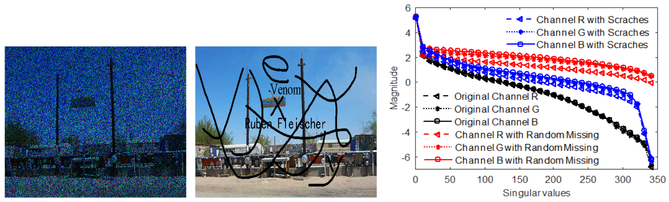
Fig. 2. Gradient magnitude histogram of sample image (a). (b)-(h): The gradient distributions of pixel matrices R, G, B, inter-channels, and unfolded modes 1, 2, 3.

values in all the three RGB channels and modes 1 and 2 is at least 341. However, this number reduces to 3 in the mode 3 unfolded matrix, so we plot the singular value distribution of mode 3 individually (Fig. 1c) for clarity. According to this figure, it is obvious that the low-rank structures are much different between the spatial and channel modes, which means the sparsity in terms of singular values is quite different. Specifically, the matrices unfolded by two spatial modes (modes 1 and 2) are approximately low-rank, but mode 3 that unfolded by channels is nearly full rank, i.e., the rank is 3. Therefore, imposing the LR constraint on all the three modes as in [11] and [35] may not be a reasonable way. Though parameters $\beta_i, i = 1, 2, 3$ can be added to weight different contributions of each mode, this leads to a tricky issue of model selection. In Fig. 1b, we can further confirm that each pixel matrix presents even lower rank structure than the spatial modes. Moreover, all pixel matrices (RGB) share very close singular value distributions to each other. These observations motivate us to apply LR constraint for each channel without weight factor instead of for unfolded mode with weight factor β_i, which will greedily reduce the computational cost and save the exhausting parameter search.

Fig. 2 plots the gradient histogram of another RGB image. Similar as the results in Fig. 1, our first observation is that the gradient distributions that computed by intra-channels are closer to each other and more compact than that by inter-channels and unfolded modes. This again motivates us applying TV constraint for pixel matrices instead of for unfolded tensor modes. Together with the discussion given around Fig. 1, we recast the multichannel completion problem as

$$\min_{\boldsymbol{X}} \frac{1}{2} \|P_\Omega(\boldsymbol{X} - \boldsymbol{M})\|^2 + \beta \sum_{i=1}^{3} \|\boldsymbol{X}^{(i)}\|_* + \lambda \|D\boldsymbol{X}\|_1, \tag{10}$$

where $\|D\boldsymbol{X}\|_1 = \sum_{k=1}^{3} \sum_{ij} |\boldsymbol{X}^{(k)}(i+1, j) - \boldsymbol{X}^{(k)}(i, j)| + |\boldsymbol{X}^{(k)}(i, j+1) - \boldsymbol{X}^{(k)}(i, j)|$ is the finite-difference operator with periodic boundary conditions [42]. Note that the TV regularizer follows an anisotropic version. We use it for two causes: it is (i) easier to implement in optimization; and (ii) more flexible to impose piecewise smoothness on different channels.



(a) Random missing (b) Scratch and text (c) Singular value distributions

Fig. 3. The variations of singular value distributions under (a) random pixels missing and (b) scratch and text corruption.

The constraints used in Eq.(10) are in convex form, which guarantees a tractable solution but may deviates far from the intrinsic sparsity priors of singular values and gradients. Fig. 3 further shows the variations of singular values under random pixel missing and scratch corruption. In Fig. 3a, the missing rates for channels R, G, and B, are 40%, 60%, and 80%, respectively. From Fig. 3c, we can see that the corruption mainly destroys the rank distribution in smaller singular values, while keeps the larger values close to the original structure. For this reason, we consider borrowing the truncated nuclear norm (TNN) [21] [22], which punishes only the smaller singular values, to our LR constrained problem. Concretely, for any matrix $\boldsymbol{Z}$ with n singular values, TNN is defined as:

$$\|\boldsymbol{Z}\|_r = \sum_{i=r+1}^{n} \sigma_i(\boldsymbol{Z}), \tag{11}$$

in which the largest r nonzero singular values are left free in implementation.

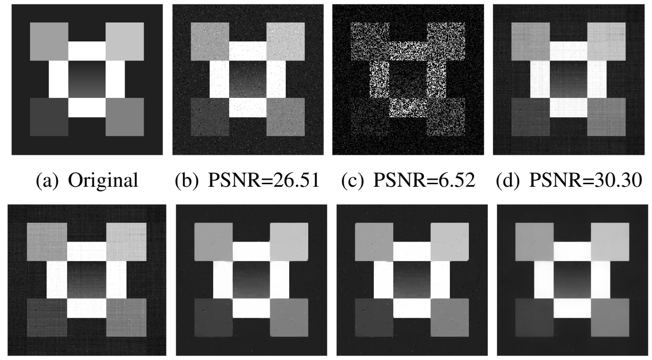
Due to the loose approximation of l_1 norm to l_0 norm, TV constraint may over smooth the real edges of natural images. Many approaches resort to l_0 gradient minimization for better results. In terms of depth images, [24] claims that both gradient magnitudes 0 and 1 should be maintained to characterize the property of low gradient. However, from Fig. 2b, 2c, and 2d, we observe that besides 0 and 1, a non-ignorable part of magnitudes owns the values larger than 1. This observation inspires us to employ l_p norm with $0 < p \leq 1$ for bridging the gap between TV norm and l_0 gradient. Hence, model (10) can finally be recast into

$$\min_{\boldsymbol{X}} \frac{1}{2}\|P_\Omega(\boldsymbol{X}-\boldsymbol{M})\|^2 + \beta\sum_{i=1}^{3}\|\boldsymbol{X}^{(i)}\|_r + \lambda\|D\boldsymbol{X}\|_p^p, \tag{12}$$

where the term $\|D\boldsymbol{X}\|_p^p$ can be named as total p variation. The cost function in (12) comprises of three terms, where the first term models the loss fidelity, and the second and third terms are both nonconvex and nonsmooth regularizers for inducing sparsity property in terms of singular values and gradients, respectively. Following [43], several assumptions can be assured for model (12).

A 1. The loss fidelity $l(\boldsymbol{X}) = \frac{1}{2}\|P_\Omega(\boldsymbol{X}-\boldsymbol{M})\|^2$ is a smooth convex function with Lipschitz gradient, and the Lipschitz constant L can be estimated as:

$$L = \sup \frac{\|\nabla l(\boldsymbol{X}_1)-\nabla l(\boldsymbol{X}_2)\|_2}{\|\boldsymbol{X}_1-\boldsymbol{X}_2\|_2} = \|\boldsymbol{Q}\|_2.$$



(a) Original (b) PSNR=26.51 (c) PSNR=6.52 (d) PSNR=30.30
(e) PSNR=32.05 (f) PSNR=35.30 (g) PSNR=36.46 (h) PSNR=37.49

Fig. 4. Visual comparison of different regularizers on a toy example. (a) Original image; (b) Noisy image with standard variance 30; (c) Observed image with both noisy corruption and 60% random pixels missing; (d)-(h) recovered images by NNM, TNN, TV, TpV, and TNN plus TpV, respectively.

A 2. The two property constraints $f_1(\boldsymbol{X}) = \beta\sum_{i=1}^{3}\|\boldsymbol{X}_{(i)}\|_r$ and $f_2(D\boldsymbol{X}) = \lambda\|D\boldsymbol{X}\|_p^p$ are proper closed functions, and they are continuous in their closed domains with $\mathrm{dom}f_1 \cap \mathrm{dom}f_2 \neq \varnothing$.

A 3. The function $l + f_1$ is level-bounded.

Based on the aforementioned choice of different regularizers, a toy example of grey image inpainting is graphically illustrated in Fig. 4, from which we have the following observations. First, by carefully comparing Fig. 4d with Fig. 4e (or Fig. 4f with Fig. 4g), we can confirm that the nonconvex constraint can better deals with edge preservation (or structural smoothness) than the convex counterpart; Second, while the LR prior promotes clearer edges than the TV prior, the latter reduces more artifacts in smooth regions than the former one does; Finally, the combination of TNN and TpV generates the best result, which improves the piecewise smoothness and simultaneously preserves sharp edges.

IV. OPTIMIZATION SCHEME

One advantage of model (12) over that in [11] and [35] is our matrices $\boldsymbol{X}^{(k)}$, $k = 1, 2, 3$ possess different entries with each other, which means the variables are not interdependent and holds the possibility to be minimized without any splitting schemes. However, the nonsmooth nonconvex constraint $f(\boldsymbol{X}) = f_1(\boldsymbol{X}) + f_2(\boldsymbol{X})$ can be complicated in practice and handling it directly is challenging.

On one hand, when one of the two parameters β and λ equals 0, then some standard first-order approaches such as accelerated proximal gradient (APG) or its variants [22] can be applied to model (12). However, practically β and λ are always set as positive values to induce both low-rank and low-gradient properties and the linear mapping D makes the optimization even tougher, so the APG-based algorithms cannot be applied efficiently, because the proximal mapping of $f(\boldsymbol{X})$ is difficult to compute.

On the other hand, when we set $r = 0$ and $p = 1$, one alternative approach for solving (12) is the ADMM algorithm; see, for example, [11][18][21][35]. This algorithm can be applied by suitably introducing slack variables that split the problem into several sub-optimization problems, resulting in

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separately computing the proximal mappings of f_1 and f_2, as well as iteratively updating of several dual variables. However, for $r > 0$ and $0 < p < 1$, it is known that ADMM does not necessarily converge due to the nonconvexity of model (12).

In this section, we propose a new scheme that is ready to take advantage of the efficient proximal mapping computations and has convergence guarantee under mild assumption. We start by introducing some preliminary results. Then our optimization scheme is presented followed by the detailed convergence analysis. Note that in the following Preliminaries subsection, we denote the symbols in vector form for simplicity. Their extension from vector to matrix/tensor is straightforward.

A. Preliminaries

For any function h: $\Re^n \to \Re \cup \{\infty\}$, it is said to be proper if dom $h = \{\boldsymbol{x} : h(\boldsymbol{x}) < \infty\} \neq \varnothing$. Such function is further said to be closed if it is lower semicontiguous. The limiting and horizon subdifferentials of h at $\boldsymbol{x}$ $\in$dom h with regarding to any sequence number t can be respectively defined as [44]:

(D1) $\partial h(\boldsymbol{x}) = \{\boldsymbol{u} : \exists \boldsymbol{u}^t \to \boldsymbol{u}, \boldsymbol{x}^t \xrightarrow{h} \boldsymbol{x}$ with $\liminf\limits_{\boldsymbol{v}\to\boldsymbol{x}^t,\boldsymbol{v}\neq\boldsymbol{x}^t} \frac{h(\boldsymbol{v})-h(\boldsymbol{x}^t)-<\boldsymbol{u}^t,\boldsymbol{v}-\boldsymbol{x}^t>}{\|\boldsymbol{v}-\boldsymbol{x}^t\|} \geq 0\}$;

(D2) $\partial^h h(\boldsymbol{x}) = \{\boldsymbol{u} : \exists \alpha_t \downarrow 0, \alpha_t \boldsymbol{u}^t \to \boldsymbol{u}, \boldsymbol{x}^t \xrightarrow{h} \boldsymbol{x}$ with $\liminf\limits_{\boldsymbol{v}\to\boldsymbol{x}^t,\boldsymbol{v}\neq\boldsymbol{x}^t} \frac{h(\boldsymbol{v})-h(\boldsymbol{x}^t)-<\boldsymbol{u}^t,\boldsymbol{v}-\boldsymbol{x}^t>}{\|\boldsymbol{v}-\boldsymbol{x}^t\|} \geq 0\}$.

where $\boldsymbol{x}^t \xrightarrow{h} \boldsymbol{x}$ denotes simultaneously $\boldsymbol{x}^t \to \boldsymbol{x}$ and $h(\boldsymbol{x}^t) \to h(\boldsymbol{x})$. Moreover, these two subdifferentials have the robustness nature such as:

$$\begin{aligned}&\{\boldsymbol{u} : \exists \boldsymbol{u}^t \to \boldsymbol{u}, \boldsymbol{x}^t \xrightarrow{h} \boldsymbol{x} \text{ with } \boldsymbol{u}^t \in \partial h(\boldsymbol{x}^t)\} \subseteq \partial h(\boldsymbol{x}),\\ &\{\boldsymbol{u} : \exists \alpha_t \downarrow 0, \alpha_t \boldsymbol{u}^t \to \boldsymbol{u}, \boldsymbol{x}^t \xrightarrow{h} \boldsymbol{x} \text{ with } \boldsymbol{u}^t \in \partial h(\boldsymbol{x}^t)\} \subseteq \partial^h h(\boldsymbol{x}).\end{aligned} \tag{13}$$

For any proper and closed function h and given $\gamma > 0$, the Moreau envelope is defined as:

$$m_\gamma h(\boldsymbol{x}) = \inf_{\boldsymbol{y}} \left\{ \tfrac{1}{2\gamma} \|\boldsymbol{x} - \boldsymbol{y}\|^2 + h(\boldsymbol{y}) \right\},$$

for which the infimum can be attained by the so-called proximal mapping of γh at $\boldsymbol{x}$, i.e.,

$$\text{prox}_{\gamma h}(\boldsymbol{x}) = \arg\min_{\boldsymbol{u}\in\Re^n} \left\{ \tfrac{1}{2} \|\boldsymbol{x} - \boldsymbol{u}\|^2 + \gamma h(\boldsymbol{u}) \right\};$$

This set is often nonempty since the function h is proper closed and bounded below [44]. Let $\boldsymbol{\zeta}_\gamma \in \text{prox}_{\gamma h}(\boldsymbol{x})$, then from [44] one can get that

$$\boldsymbol{x} - \boldsymbol{\zeta}_\gamma \in \gamma \partial h(\boldsymbol{\zeta}_\gamma). \tag{14}$$

Moreover, the following two propositions can be easily derived. For self-containedness, we provide their proofs in the Appendix.

Proposition 1. For any nonnegative function h (as a constraint) and $\gamma \geq 0$, the function m_γ satisfies

$$0 \leq m_\gamma h(\boldsymbol{x}) \leq h(\boldsymbol{x}), \forall \boldsymbol{x} \in \Re^n, \tag{15}$$

Proposition 2. Suppose function h is proper and closed with $\inf h > -\infty$, let t be any sequence number and $\boldsymbol{x}^* \in$dom h, then given $\boldsymbol{x}^t \to \boldsymbol{x}^*$, $\gamma_t \downarrow 0$ and $\boldsymbol{\zeta}^t \in \text{prox}_{\gamma_t h}(\boldsymbol{x}^t)$, we have $\boldsymbol{\zeta}^t \in$dom h and $\boldsymbol{\zeta}^t \to \boldsymbol{x}^*$.

B. Moreau Approximated DC algorithm

In this subsection, we adapt a smoothing scheme for solving the nonconvex nonsmooth problem (12). In a broad way, we minimize the surrogate function

$$F_\gamma(\boldsymbol{X}) = \frac{1}{2} \|P_\Omega(\boldsymbol{X} - \boldsymbol{M})\|^2 + \beta \sum_{i=1}^{3} \left\|\boldsymbol{X}^{(i)}\right\|_r + \lambda m_\gamma \|D\boldsymbol{X}\|_p^p, \tag{16}$$

approximately and then update $\boldsymbol{X}$ and γ in each iteration, where $m_\gamma \parallel D\boldsymbol{X} \parallel_p^p$ is the Moreau envelope of f_2. The key ingredient in our scheme is the simple consideration that for f_2 and any $\gamma > 0$,

$$\begin{aligned} m_\gamma f_2(D\boldsymbol{X}) =& \frac{1}{2\gamma} \|D\boldsymbol{X}\|^2 \\ &- \underbrace{\sup_{\boldsymbol{Y}} \{ \frac{1}{\gamma} tr((D\boldsymbol{X})^T \boldsymbol{Y}) - \frac{1}{2\gamma} \|\boldsymbol{Y}\|^2 - f_2(\boldsymbol{Y}) \}}_{P_{\gamma f_2}(D\boldsymbol{X})}, \end{aligned} \tag{17}$$

where $P_{\gamma f_2}$ is finite-valued as being the supreme of a convex, continuous, and affine function. Then, we have

$$\begin{aligned} \frac{1}{\gamma}\text{prox}_{\gamma f_2}(D\boldsymbol{X}) \subseteq \partial P_{\gamma f_2}(D\boldsymbol{X}) \Rightarrow \frac{1}{\gamma} D^{\mathrm{T}} \text{prox}_{\gamma f_2}(D\boldsymbol{X}) \\ \subseteq D^{\mathrm{T}} \text{prox}_{\gamma f_2}(D\boldsymbol{X}) = \partial (P_{\gamma f_2} \circ D)(\boldsymbol{X}), \end{aligned} \tag{18}$$

where D^{T} is the inverse operator of D [42], the equality follows from [44] since that $P_{\gamma f_2}$ is convex continuous.

Thus, problem (16) is the sum of a Lipschitz smooth function l, a nonconvex nonsmooth constraint f_1, and a continuous DC decomposition Eq.(17). Thanks to Eq.(17), the subgradient corresponding to the concave part of Eq.(16), i.e., $-P_{\gamma f_2}(D\boldsymbol{X})$, can be easily computed. Consequently, certain DC-based algorithms [45][46][47] with majorization techniques can be suitably applied. In our implementation, we employ the proximal DC algorithm with extrapolation (DCA_e) [45] by recasting Eq.(16) as follows.

$$\begin{aligned} F_\gamma(\boldsymbol{X}) = & \underbrace{\frac{1}{2} \|P_\Omega(\boldsymbol{X} - \boldsymbol{M})\|^2 + \frac{\lambda}{2\gamma} \|D\boldsymbol{X}\|^2}_{g(\boldsymbol{X})} + \underbrace{\beta \sum_{i=1}^{3} \left\|\boldsymbol{X}^{(i)}\right\|_r}_{f_1(\boldsymbol{X})} \\ & - \underbrace{\lambda P_{\gamma f_2}(D\boldsymbol{X})}_{P_2(\boldsymbol{X})}, \end{aligned} \tag{19}$$

Due to the assumption A3 and the non-negativity of m_γ, it is obvious that F_γ is level-bounded and continuous in its domain. Then all the conditions required by DCA_e [45] are satisfied and the algorithm can be applied for our model by initializing at any 3-order tensor $\boldsymbol{X}^0 \in \Re^{I_1 \times I_2 \times 3}$.

We now describe our scheme for solving (12) in Algorithm 1. We call this method the Moreau approximated difference-of-convex algorithm (MADC). Note that in the inner loop, i.e., Step 2, we also adopt the extrapolation trick as DCA_e dose for acceleration. In detail, as for satisfying the condition $\eta_{l_t}^t \in [0, 1)$ and $\sup_{l_t} \eta_{l_t}^t < 1$, we start with $\omega_{-1} = \omega_0 = 1$, and recursively assigns

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$\eta_{l_t}^t = \frac{\omega_{l_t-1}-1}{\omega_{l_t}}$ with $\omega_{l_t+1} = \frac{1+\sqrt{1+4\omega_{l_t}^2}}{2}$.

Algorithm 1 The MADC for color image completion.

Input: Given the observed image $\boldsymbol{X}^{ob}$, pick random tensors $\boldsymbol{X}^{-1} = \boldsymbol{X}^0 \in \Re^{I_1\times I_2\times 3}$ and two sequences of positive values $\varepsilon_t \downarrow 0$, $\gamma_t \downarrow 0$ with $t \in \{0,1,\ldots\}$; Initialize $t = 0$.

Output: $\boldsymbol{X}^t$.

Step1. when $F_{\gamma_t}(\boldsymbol{X}^t) \leqslant F_{\gamma_t}(\boldsymbol{X}^{ob})$, set $\boldsymbol{X}_0^t = \boldsymbol{X}^t$. Otherwise, set $\boldsymbol{X}_0^t = \boldsymbol{X}^{ob}$.

Step2. **for** $l_t = 0,1,2,\ldots$

Pick $\eta_{l_t}^t \in [0,1)$ with $\sup_{l_t} \eta_{l_t}^t < 1$.

Compute $\boldsymbol{\phi}_{l_t}^t \in \partial P_2(\boldsymbol{X}_{l_t}^t)$ via (18) and set

$$\boldsymbol{Y}^t = \boldsymbol{X}_{l_t}^t + \eta_{l_t}^t(\boldsymbol{X}_{l_t}^t - \boldsymbol{X}_{l_t-1}^t) \tag{20}$$

$$\begin{aligned}\boldsymbol{X}_{l_t+1}^{t(i)} = \text{argmin}_{\boldsymbol{Y}^{(i)}}\{&< \nabla g(\boldsymbol{Y}^{t(i)}) - \boldsymbol{\phi}_{l_t}^{t(i)}, \boldsymbol{Y}^{(i)} > \\ &+(L/2)\|\boldsymbol{Y}^{(i)} - \boldsymbol{Y}^{t(i)}\|^2 + \|\boldsymbol{Y}^{(i)}\|_r\}, i = 1,2,3.\end{aligned} \tag{21}$$

if

$$\begin{aligned}&\text{dist}(0, \nabla l(\boldsymbol{X}_{l_t}^t) + \partial f_1(\boldsymbol{X}_{l_t+1}^t) + (1/\gamma)D^{\mathrm{T}}[\lambda D\boldsymbol{X}_{l_t}^t - \\ &\text{prox}_{\gamma_t f_2}(D\boldsymbol{X}_{l_t}^t)]) \le \varepsilon_t, \left\|\boldsymbol{X}_{l_t+1}^t - \boldsymbol{X}_{l_t}^t\right\| \le \varepsilon_t, \\ &\text{and } F_{\gamma_t}(\boldsymbol{X}_{l_t}^t) \le F_{\gamma_t}(\boldsymbol{X}_0^t) \text{ , \textbf{then break}.}\end{aligned} \tag{22}$$

end for

Step3. Update $\boldsymbol{X}^{t+1} = \boldsymbol{X}_{l_t}^t$ and $t = t+1$. Go to **Step1**.

C. Analysis of Convergence and Complexity

In this subsection, we first comment on the computational cost of Algorithm 1. Then we prove the convergence of MADC under mild assumption.

The main computational complexity of Algorithm l lies in two proximal operators with identity mapping, i.e., prox$(\|\cdot\|_p^p)$ and prox$(\|\cdot\|_r)$, where the former is used for Moreau approximation and the latter is used to solve (21). These two operators can be easily computed following Theorem 1 and Theorem 2, respectively. For color image completion with three channels, the cost of $\text{prox}_{\tau p}$ and $\text{prox}_{\tau r}$ in each iteration are $O(3I_1I_2)$ and $O(3I_1I_2\text{min}(I_1,I_2))$, respectively.

Theorem 1. [46]. For any $\tau > 0$ and $\boldsymbol{y} \in \Re^n$, define the proximal operator with l_p constraint as

$$\text{prox}_{\tau p}(\boldsymbol{y}) = \underset{\boldsymbol{x}\in\Re^n}{\arg\min}\left\{\tfrac{1}{2}\left\|\boldsymbol{x}-\boldsymbol{y}\right\|^2 + \tau\left\|\boldsymbol{x}\right\|_p^p\right\}.$$

When p=1, $\text{prox}_{\tau 1}(\boldsymbol{y})$ is the well-known soft-thresholding operator that has closed form solution as $\text{prox}_{\tau 1}(\boldsymbol{y})_i = S_\tau(\boldsymbol{y})_i = \text{sign}(y_i)\text{max}(|y_i| - \tau, 0)$, with $i = 1,2,\ldots,n$. When $0 < p < 1$, this operator can be solved as

$$\text{prox}_{\tau p}(\boldsymbol{y})_i = \begin{cases} 0, & |y_i| < \vartheta + p\tau\vartheta^{p-1} \\ \{0, sign(y_i)\vartheta\}, & |y_i| = \vartheta + p\tau\vartheta^{p-1} \\ sign(y_i)z_i & |y_i| > \vartheta + p\tau\vartheta^{p-1} \end{cases};$$

for $i = 1,2,\ldots,n$, where $\vartheta = [2\tau(1-p)]^{1/(2-p)}$, z_i is the solution of $h(z) = pz^{p-1} + (z - |y_i|)/\tau = 0$ over the span$(\vartheta, |y_i|)$.

Theorem 2. [47]. For any $\tau > 0$ and $\boldsymbol{Y} \in \Re^{I_1\times I_2}$, define the proximal operator with TNN constraint as

$$\text{prox}_{\tau r}(\boldsymbol{Y}) = \underset{\boldsymbol{X}\in\Re^{I_1\times I_2}}{\arg\min}\left\{\tfrac{1}{2}\left\|\boldsymbol{X}-\boldsymbol{Y}\right\|^2 + \tau\left\|\boldsymbol{X}\right\|_r\right\}.$$

Then, its optimal solution can be computed by

$$\text{prox}_{\tau r}(\boldsymbol{Y}) = \boldsymbol{U}(\boldsymbol{\Sigma}_1 + S_\tau[\boldsymbol{\Sigma}_2])\boldsymbol{V}^T,$$

where $\boldsymbol{U\Sigma V}^T$ is the SVD of $\boldsymbol{Y}$, $\boldsymbol{\Sigma}_1$ and $\boldsymbol{\Sigma}_2$ contain the r largest singular values and the last $\min(I_1, I_2) - r$ singular values, respectively; $S_\tau[\cdot]$ is with the same definition as in Theorem 1.

As discussed in ([45], Theorem 4.2), F_{γ_t} can be efficiently minimized so that condition (22) will be satisfied after several iterations. This leaves us to prove the convergence result for outer loop of MADC. We now state assumption A4 and present two propositions (the proofs are given in the Appendix) followed by Theorem 3 to prove the convergence of Algorithm 1. Note that assumption A4 is a mild condition for nonconvex nonsmooth optimization issues; see ([44], Corollary 10.9). It can be satisfied for our model since that D is a linear mapping and domf_2 is continuous.

A 4. Given $\boldsymbol{X}^*$ as an accumulation point of sequence $\{\boldsymbol{X}^t\}$, we have

$$\boldsymbol{Y}_1 + D^{\mathrm{T}}\boldsymbol{Y}_2 = 0 \text{ and } \boldsymbol{Y}_1 \in \partial^h f_1(\boldsymbol{X}^*), \boldsymbol{Y}_2 \in \partial^h f_2(D\boldsymbol{X}^*) \Rightarrow \boldsymbol{Y}_2 = 0.$$

Proposition 3. Let $\boldsymbol{X}^*$ be the accumulation point of sequence $\{\boldsymbol{X}^t\}$ and $\varGamma$ be an index set with $\lim_{t\in\varGamma}\boldsymbol{X}^t = \boldsymbol{X}^*$, given $\boldsymbol{\mu}^t \in \partial f_1(\boldsymbol{X}_{l_t}^t)$, $\boldsymbol{\zeta}^t \in \text{prox}_{\gamma_{t-1}f_2}(D\boldsymbol{X}^t)$, and $r_t = \|\boldsymbol{\mu}^t\| + (1/\gamma_{t-1})\|D^{\mathrm{T}}(\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t)\|$, then $\{\varPsi^t\}_{t\in\varGamma}$ with $\varPsi^t = \|(\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t)/(\gamma_{t-1}r_t)\|$ is bounded.

Proposition 4. With the same definitions as in Proposition 3, sequence $\{r_t\}_{t\in\varGamma}$ is bounded.

Theorem 3. (Convergence of MADC). The sequence $\{\boldsymbol{X}^t\}$ computed by applying MADC to model (12) is bounded. Under assumption A4, it holds that $\boldsymbol{X}^* \in \text{dom}f_1 \cap D^{-1}\text{dom}f_2$, and $\boldsymbol{X}^*$ is a stationary point of model (12), i.e.,

$$\mathbf{0} \in \nabla l(\boldsymbol{X}^*) + \partial f_1(\boldsymbol{X}^*) + D^{\mathrm{T}}\partial f_2(D\boldsymbol{X}^*) \tag{23}$$

Proof. Using the last relation in Eq.(22), we have

$$l(\boldsymbol{X}^t) + f_1(\boldsymbol{X}^t) \le F_{\gamma_{t-1}}(\boldsymbol{X}^t) \le F_{\gamma_{t-1}}(\boldsymbol{X}^{ob}) \le F(\boldsymbol{X}^{ob}) \tag{24}$$

where the first inequality follows from the non-negativity of f_2 and the last inequality follows from Proposition 1. This together with assumption A3 assure that $\{\boldsymbol{X}^t\}$ is bounded. Then, using the definition $\lim_{t\in\varGamma}\boldsymbol{X}^t = \boldsymbol{X}^*$, (24), and the closeness of $l + f_1$, we see

$$l(\boldsymbol{X}^*) + f_1(\boldsymbol{X}^*) \le \liminf_{t\in\varGamma} l(\boldsymbol{X}^t) + f_1(\boldsymbol{X}^t) \le F(\boldsymbol{X}^{ob}) < \infty,$$

which gives out $\boldsymbol{X}^* \in \text{dom}f_1$. Besides, it holds that

$$0 \le \tfrac{1}{2}\text{dist}^2(D\boldsymbol{X}, \text{dom}f_2) \le \inf_{\boldsymbol{Y}}\{\tfrac{1}{2}\|D\boldsymbol{X} - \boldsymbol{Y}\|^2 + \gamma_{t-1}f_2(\boldsymbol{Y})\} = \gamma_{t-1}m_{t-1}f_2(D\boldsymbol{X}),$$

where the second inequality follows from the definition of dist$(\cdot)$ [42]. With (24), this can be further extended into

$$\tfrac{1}{2\gamma_{t-1}}\text{dist}^2(D\boldsymbol{X}, \text{dom}f_2) \le m\gamma_{t-1}f_2(D\boldsymbol{X}) \le F_{\gamma_{t-1}}(\boldsymbol{X}^t) \le F(\boldsymbol{X}^{ob}).$$

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With the fact that $\gamma_t \downarrow 0$, one can conclude $\mathrm{dist}^2(D\boldsymbol{X}^*, \mathrm{dom} f_2) \leq 0$ and hence $D\boldsymbol{X}^* \in \mathrm{dom} f_2$ due to the closeness of dom f_2.

Next, we prove (23) under assumption A4. Without loss of generality, for certain $\tilde{\boldsymbol{\mu}}^*$ and $\tilde{\boldsymbol{\chi}}^*$, we may assume

$$\lim_{t\in\Gamma} \boldsymbol{\mu}^t = \tilde{\boldsymbol{\mu}}^* \text{ and } \lim_{t\in\Gamma} \frac{1}{\gamma_{t-1}} D^{\mathrm{T}}(\lambda D X^t - \boldsymbol{\zeta}^t) = \tilde{\boldsymbol{\chi}}^* \tag{25}$$

since that $\{r_t\}_{t\in\Gamma}$ is bounded from Proposition 4. Then with (13), we get

$$\tilde{\boldsymbol{\mu}}^* \in \partial f_1(\boldsymbol{X}^*) \tag{26}$$

On the other hand, following the similar statements of Proposition 3, we can assure that $\{\Lambda^t\}_{t\in\Gamma}$ with $\Lambda^t = \|(\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t)/\gamma_{t-1}\|$ is bounded. Then $\lim_{t\in\Gamma}(\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t)/\gamma_{t-1}$ exists. This together with (25) leads to

$$\begin{aligned}\tilde{\boldsymbol{\chi}}^* =& D^{\mathrm{T}} \lim_{t\in\Gamma} \frac{\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t}{\gamma_{t-1}} \in D^{\mathrm{T}}\{\lim_{t\in\Gamma} \boldsymbol{u}^t : \boldsymbol{u}^t \in \partial f_2(\zeta^t)\} \\ &\subseteq D^{\mathrm{T}} \partial f_2(D\boldsymbol{X}^*)\end{aligned} \tag{27}$$

where the left inclusion follows from (14) and the right one follows from Proposition 2 as well as definition D2. Taking (26), (27) and (32) together along with $t \in \Gamma$, we conclude that

$$\begin{aligned}\mathbf{0} = \nabla l(\boldsymbol{X}^*) + \tilde{\boldsymbol{\mu}}_* + \tilde{\boldsymbol{\chi}}_* \in \nabla l(\boldsymbol{X}^*) + \partial P_1(\boldsymbol{X}^*) + \\ D^T \partial f_2(D\boldsymbol{X}^*),\end{aligned} \tag{28}$$

which completes the proof.

V. EXPERIMENTAL RESULTS

To validate the performance of the proposed approach, we apply it on a variety of natural images with different inpainting tasks, i.e., scratch and text removal, randomly missing pixels filling, and the completion under mixed corruption. All the used samples are shown in Fig. 5, where we select 9 images with different scenes and sizes for assessment of both effectiveness and efficiency. We compare our approach with several state-of-the-art completion algorithms, i.e., HaLR [33], PDS [11], ADMM [35], MCInd [24], and BCD [34]. Moreover, we also implement our MC problem in a 3D mode (MC3D) using ADMM scheme for a direct comparison of different optimization strategies. To make a fair comparison between algorithms, the tunable parameters for all the competing methods are manually selected to report the best performance in terms of quantitative criteria, visual assessment, and computational cost. Furthermore, the maximum iteration number m_t and the stop tolerance ε are set as 200 and 1e-5, respectively. Unless specified otherwise, the parameters r and p of our method are set as 9 and 1 at default.

We adopt the commonly used Peak Signal-to-Noise (PSNR) value to evaluate the quantitative performance of different approaches. Suppose that the total number of missing pixels is N, then the mean squared error $MSE = \sum_{i=1}^{3} e_i^2/3N$, and PSNR can be obtained by $10\log_{10}(255^2/MES)$. All experiments are performed on a program platform with Intel Core i7-8500U 1.8GHz and 8GRAM. The source code for our method and the used images are offered as the supplemental material and will be released online.

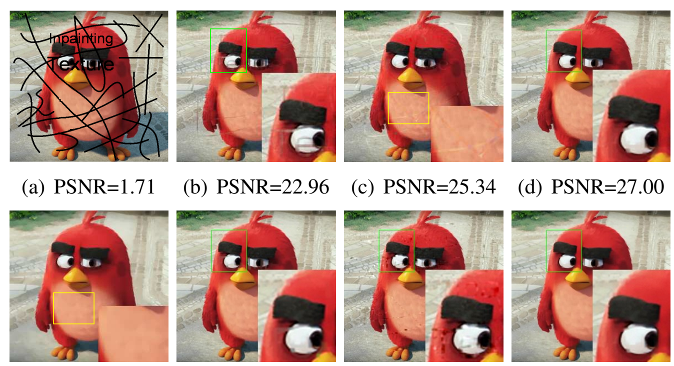
(1) 341×600 (2) 300×300 (3) 512×512 (4) 500×500 (5) 512×512

(6) 382×656 (7) 1300×975 (8) 643×950 (9) 408×612

Fig. 5. The images (1-9) used in the experiments.

A. Experiments on Scratch and Text Removal

Scratch and/or text removal is a practical problem in many image processing applications. For each given sample in Fig. 5, we scribble some texts and curves on it and regard the corresponding pixels as missing entries. So, this problem can be considered as a general completion task. In Table I, we quantify the recovery results in terms of PSNR from all competing methods, where I1, . . . , I9 denote image 1, . . . , image 9, respectively, for simplicity. Moreover, the visual comparisons of different completion algorithms on image 5 and image 9 are shown in Fig. 6 and Fig. 7, respectively.



(a) PSNR=1.71 (b) PSNR=22.96 (c) PSNR=25.34 (d) PSNR=27.00

(e) PSNR=26.64 (f) PSNR=27.14 (g) PSNR=23.84 (h) PSNR=28.00

Fig. 6. Recovery results of (b) HaLR, (c) PDS, (d) ADMM, (e) MCInd, (f) MC3D, (g) BCD, and (h) MADC methods on (a) the scratch and text corrupted image 5.

From Table I, it can be seen that BCD ranks bottom in most of the test images. We attribute this to the improper imposing of TV regularization on the factor matrices for inpainting problems. Interestingly, HaLR, which neglects the local smooth and piecewise property of natural images, achieves appealing performance in many tests, e.g., images 6, and 7. This is because these images contain lots of redundant information, especially for image 7, which are intensively utilized in HaLR with the form of low-rank property. The remaining 5 competing methods all jointly adopt low-rank and TV property for the multichannel completion problems. Their variations mainly lie in the optimization scheme and the practical selection of matrix or tensor operations. We can see that our method achieves the best PSNR in most samples except for image 7 and image 8. For these two, MADC ranks second and third, respectively. On average of all the 9 images, MADC has the highest PSNR value 22.50, and the gains over HaLR, PDS, ADMM, MCInd, MC3D, and BCD are 2.37dB, 1.61dB, 0.82dB, 3.423dB, 1.76dB, and 9.59dB, respectively. The excellent performance is attributed to the joint optimization of different channels and the enhanced low-rank and TV constraints imposed on the recovered 3D images.

The results of Figs. 6 and 7 also show the improvements of our method over the others. In Fig. 6, the recovered images from HaLR and PDS still have much residual shadow of the scratches. For BCD, it produces many disturbing artefacts. Although MCInd fills most of the missing entries with correct pixels, it oversmoothes the texture on the angry bird, especially on its belly. ADMM and MC3D achieve very similar results to that of our method, but still suffer from some undesirable artifacts, such as more residual shadow in the right eye of the angry bird. In Fig. 7, HaLR, PDS, MCInd, MC3D, and BCD clearly generate worse results compared to ours. Although ADMM also produces pleasing result visually, it still performs poorer than MADC quantitatively.

(a) PSNR=1.36 (b) PSNR=24.65 (c) PSNR=26.26 (d) PSNR=27.78

(e) PSNR=12.05 (f) PSNR=20.18 (g) PSNR=8.63 (h) PSNR=27.94

Fig. 7. Recovery results of (b) HaLR, (c) PDS, (d) ADMM, (e) MCInd, (f) MC3D, (g) BCD, and (h) MADC methods on (a) the scratch and text corrupted image 9.

B. Experiments on Random Pixels Missing

Due to coding or transmission issues, partial pixels of the images may be randomly missing. In this subsection, we test the performance of the competing algorithms on recovering random missing pixels, and report the experimental results for the used images with different observed percentage. Fig. 8 presents the recovery results on image 1 with 60%, 70%, and 80% random missing pixels in channels R, G, and B, respectively. From this figure, our first observation is that MCInd and BCD both generate poorer perceptual results. The independent treatment of different channels is the main reason for the failure of MCInd. Although the performance may be slightly improved by careful tuning of the parameters for each channel, it is a tricky work and is unreasonable for practical application. The visual quality of recovered images from other competitors appears similar at first glance. However, in the magnified views of the red box, the standing poles produced by PDS, ADMM, and MC3D are all in segmented status. The result generated by HaLR retains a perfect standing pole, but the whole recovered image is much blurrier than the proposed method. Overall, our method MADC not only fills the desirable pixels, but also preserves sharper edges and finer details, showing better visual quality than the other competing methods.

(a) PSNR=4.223 (b) PSNR=22.17 (c) PSNR=21.08 (d) PSNR=21.52

(e) PSNR=14.60 (f) PSNR=22.30 (g) PSNR=13.24 (h) PSNR=22.78

Fig. 8. Recovery results of (b) HaLR, (c) PDS, (d) ADMM, (e) MCInd, (f) MC3D, (g) BCD, and (h) MADC methods on (a) the 60% random missing image 1.

For a comprehensive comparison, we further independently pose certain pixels missing for each channel and then evaluate the performance of our method with all the 9 test images. Fig. 9 shows the PSNR values versus different missing rates, i.e., $50\%, \ldots, 90\%$, of all the methods, from which we can see the obvious improvement of our method over the others, especially when the missing rate is higher than 60%. Similar as the results of Table I, BCD again performs the worst on average due to the improper penalization of TV constraint on the factor matrices. The matrix based method MCInd also performs poorly in most cases due to its separated optimization for channels R, G, and B. MC3D, another matrix completion approach, outperforms MCInd consistently in all the images, which demonstrates the advantage of joint channel optimization as analysed in Section III. For HaLR, PDS and ADMM, three tensor completion approaches, their average results are quantitatively close to the ones from MC3D and slightly poorer than that from ours. This again verifies the superiority of our enhanced low-rank total variation constraint and the proposed optimization algorithm.

C. Experiments on Mixed Corruption

In this experiment, we test the performance of our method for the case of mixture missing and noise corruption. Specifically, the test images are first corrupted by additive Gaussian noise with standard variance 10, then both scratches and texts described in Subsection V.A and random missing described in Subsection V.B are further combined to degrade the test images. With the missing pixels of scratches and texts fixed, Tables II and III list the numerical results under random missing ratios being 60% and 80%, respectively. Moreover, the visual results of image 4 under 50% missing pixels are shown in Fig. 10.

Compared to the previous two experiments, our first thought is that this experiment must be the toughest one for image recovery. Surprisingly, most of the PSNR values in Tables II and III are higher than those listed in Table I. This is because all the competing approaches are designed specific

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TABLE I
RECOVERY RESULTS (PSNR: DB) BY VARIOUS COMPLETION METHODS FROM SCRATCH AND TEXT CORRUPTED IMAGES. (THE RESULTS RANK FIRST AND SECOND IN EACH IMAGE ARE HIGHLIGHTED WITH BOLDFACE AND UNDERLINE, RESPECTIVELY)

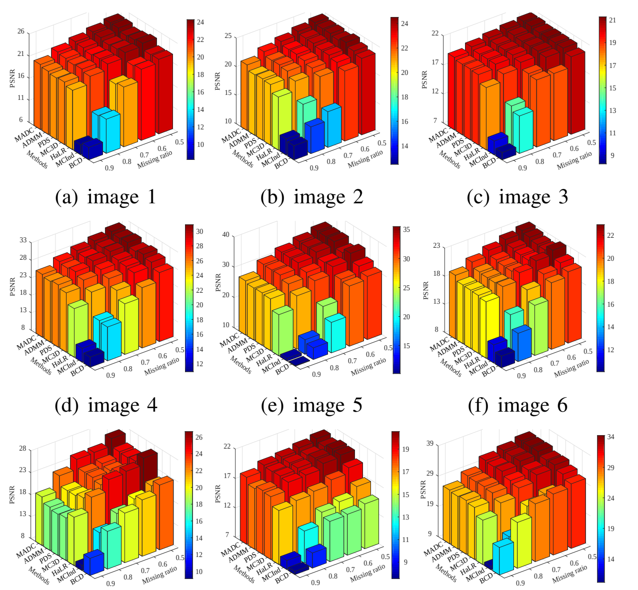
Methods	I1	I2	I3	I4	I5	I6	I7	I8	I9
HaLR	20.28	21.50	18.37	22.32	22.96	18.59	***20.76***	15.69	20.65
PDS	20.07	22.59	18.97	25.68	25.34	16.83	14.74	17.49	26.26
ADMM	21.24	22.75	19.04	25.18	27.00	17.89	16.79	17.40	27.78
MCInd	20.38	22.16	19.04	25.12	26.64	17.06	13.90	17.11	12.05
MC3D	21.06	22.61	19.13	25.17	27.14	17.65	16.18	***17.50***	20.18
BCD	10.30	19.29	14.54	15.85	23.84	9.50	7.651	6.583	8.63
MADC	***21.25***	***23.21***	***19.41***	***25.88***	***28.00***	***18.74***	20.61	17.45	***27.94***

TABLE II
RECOVERY RESULTS (PSNR: DB) FROM IMAGES SUFFERING MIXED CORRUPTION WITH MISSING RATE 60%. (THE RESULTS RANK FIRST AND SECOND IN EACH IMAGE ARE HIGHLIGHTED WITH BOLDFACE AND UNDERLINE, RESPECTIVELY)

Methods	I1	I2	I3	I4	I5	I6	I7	I8	I9
HaLR	21.87	21.83	19.71	24.75	25.13	20.66	24.32	18.11	25.57
PDS	22.13	22.91	20.22	***28.07***	29.59	20.26	21.18	19.31	28.96
ADMM	22.65	22.90	20.32	27.63	29.06	20.49	21.17	19.30	29.06
MCInd	21.36	22.24	19.86	26.70	26.83	19.29	17.75	17.91	13.06
MC3D	22.64	22.83	20.33	27.79	29.44	20.38	21.21	19.25	22.65
BCD	17.62	20.28	19.32	24.02	24.87	15.98	16.12	12.34	12.35
MADC	***22.68***	***23.23***	***20.47***	***28.07***	***29.74***	***20.73***	***24.97***	***19.52***	***29.57***

TABLE III
RECOVERY RESULTS (PSNR: DB) FROM IMAGES SUFFERING MIXED CORRUPTION WITH MISSING RATE 80%. (THE RESULTS RANK FIRST AND SECOND IN EACH IMAGE ARE HIGHLIGHTED WITH BOLDFACE AND UNDERLINE, RESPECTIVELY)

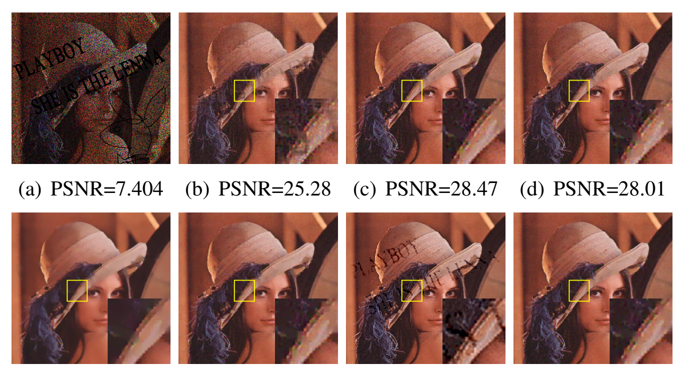
Methods	I1	I2	I3	I4	I5	I6	I7	I8	I9
HaLR	20.15	20.27	18.44	22.87	23.53	18.98	22.83	16.83	23.88
PDS	20.91	21.73	19.54	***26.70***	28.25	19.09	20.09	18.44	27.87
ADMM	21.42	21.60	19.53	26.27	28.01	19.31	20.04	18.41	27.91
MCInd	18.03	20.76	18.05	22.94	20.22	17.49	16.02	15.71	10.98
MC3D	21.29	21.52	19.57	26.18	27.91	19.11	20.09	18.33	20.87
BCD	15.55	14.78	12.67	20.56	19.91	12.13	13.12	9.42	9.90
MADC	***21.50***	***22.10***	***19.81***	26.69	***28.44***	***19.56***	***22.95***	***18.82***	***28.10***



(a) image 1 (b) image 2 (c) image 3
(d) image 4 (e) image 5 (f) image 6
(g) image 7 (h) image 8 (i) image 9

Fig. 9. Performance of competing completion algorithms for random missing problem. (a)-(i) Experimental results from image 1 to 9, respectively.

to the Gaussian distributed error, and the scratches and texts are clearly in non-Gaussian structure. With the addition of randomly missed pixels, the reconstruction residuals follow closer to the Gaussian distribution, and further lead to a better



(a) PSNR=7.404 (b) PSNR=25.28 (c) PSNR=28.47 (d) PSNR=28.01
(e) PSNR=27.45 (f) PSNR=28.26 (g) PSNR=25.29 (h) PSNR=28.51

Fig. 10. Recovery results from (b) HaLR, (c) PDS, (d) ADMM, (e) MCInd, (f) MC3D, (g) BCD, and (h) MADC on (a) the mixed corrupted image 4.

result. From these tables, we can see that MADC clearly achieves the best numerical results compared to other methods. Specifically, the average recovery performance computed from all the cells for each method can be sorted in descending order as MADC, PDS, ADMM, MC3D, HaLR, MCInd, and BCD. Taking Table II as example, the gains of our method over the others are 0.70dB, 0.71dB, 1.38dB, 1.89dB, 3.78dB, and 6.23dB, respectively. Fig. 10 also presents similar results as Tables II and III. From Fig. 10b and 10g, we can observe that clear ghost shadow still exist in part of these images. In Fig. 10e, the result recovered by MCInd is much too

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oversmoothed such that the texture of hat and hair has faded away. The visual quality of recovered images from other four approaches appears similar at first glance. However, by carefully comparing their magnified boxes, we can see that our MADC reduces more artifacts in smooth regions. The PSNR values listed at the bottom of the subfigures also confirm the visual result.

D. Computational Cost

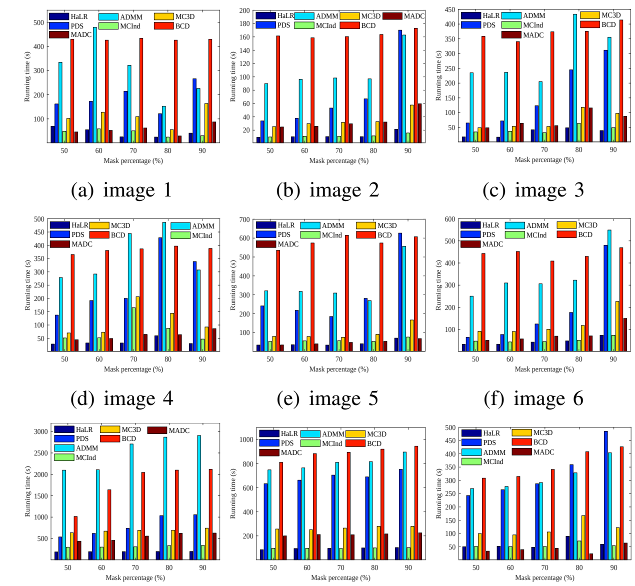
This subsection concerns the efficiency of MADC. We compare our method with other competing ones under the tasks of scratch and text removal as well as random pixel filling. The running time of all the methods when achieving the recovery results in Table I and Fig. 9 is shown in Table IV and Fig. 11, respectively. Since that BCD always runs with much more time than the other algorithms, we only show one third of its runtime in Fig. 11 for better visualization.

As depicted in Table IV and Fig. 11, the running time of HaLR is lower than other algorithms. Note that HaLR only imposes low-rank constraint on the objective variable, which costs one subproblem less than the other methods. MCInd, which resorts to matrix completion and channel-independent tricks for faster implementation, occupies the second place in terms of efficiency. Although MC3D also belongs to the matrix completion category, it implements multichannel optimization as a single unified problem, which sacrifices certain execution efficiency for a better performance (see Tables I-III). The remaining three competitors, i.e., PDS, ADMM, and BCD, are all tensor completion methods. Among them, PDS is the most efficient one due to its parsimonious use of inverse operations and restricted number of auxiliary variables. On the contrary, BCD suffers from nested strategy, i.e., the block coordinate descent scheme for outer loop as well as the alternating direction method of multipliers for inner loop, and runs the slowest in most tests. By jointly considering the quantitative values listed in the previous experiments and the results of Table IV and Fig. 11, it is evident that our MADC algorithm not only achieves the best PSNR values among all the competing methods, but also outperforms PDS, ADMM, MC3D, and BCD in terms of computational efficiency. Although the efficiency of HaLR and MCInd are higher than MADC, they lag far behind our method in terms of PSNR results.

E. Component Analysis

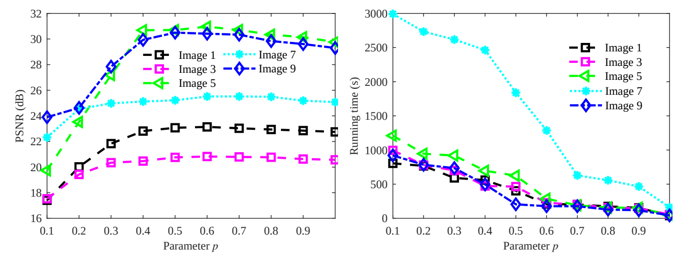
Besides the introduced jointly multichannel optimization scheme, which results in better PSNR results and appealing efficiency for our proposed method, the performance of MADC also depends on two other components. That is (i) the truncated nuclear norm relying on parameter r, and (ii) the nonconvex extension of total variation relying on parameter p. In the previous experiments, we simply fix their values for a fair comparison. Next, we evaluate the MADC algorithm using varing r and p.

Under the same experimental setting as for Fig. 10, Fig. 12 plots the PSNR and runtime curves on partial images with p traversed in $\{0.1, 0.2, \ldots, 1\}$. Note that in Fig. 12b, we scale the runtime results on image 7 to be 50% for better visualization. Our first observation is that a carefully selected parameter p can further improve the effectiveness of MADC. Generally speaking, our method performs relatively better when p lies in [0.5, 0.8]. We also observe that the smaller the p value, the more runtime required for model convergence. This is because that a smaller p leads to a more nonconvex cost function, which further results in a trickier optimization procedure. In real-world applications, $p = 1$ is a desirable value for a well balance of effectiveness and efficiency. When the user prefers to the extreme performance, then $p = 0.6$ is suggested in practice.



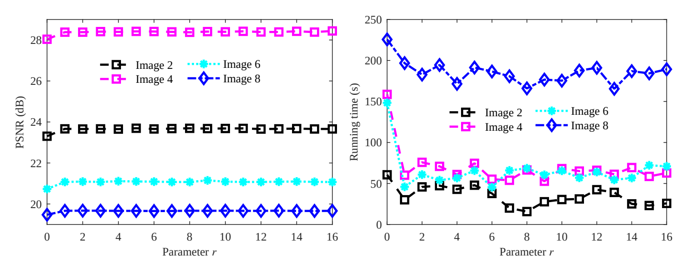
(a) image 1 (b) image 2 (c) image 3
(d) image 4 (e) image 5 (f) image 6
(g) image 7 (h) image 8 (i) image 9

Fig. 11. Running time of competing completion algorithms for random missing problem. (a)-(i) Experimental results from image 1 to 9, respectively.



(a) PSNR (b) Runtime

Fig. 12. Comparisons on (a) PSNR and (b) Runtime under different settings of parameter p.



(a) PSNR (b) Runtime

Fig. 13. Comparisons on (a) PSNR and (b) Runtime under different settings of parameter r.

TABLE IV
RUNNING TIME (IN SECONDS) OF COMPETING METHODS IN SCRATCH AND TEXT CORRUPTED IMAGES.

Methods	I1	I2	I3	I4	I5	I6	I7	I8	I9
HaLR	22.8	5.8	23.1	21.9	26.4	32.6	160.9	65.0	29.1
PDS	59.3	22.8	88.4	124.7	125.8	87.6	451.7	168.1	122.1
ADMM	113.5	53.1	245.3	237.9	197.5	189.0	1876	562.6	159.1
MCInd	28.9	9.0	55.0	63.0	46.5	44.3	353.5	103.8	38.7
MC3D	48.8	16.2	65.9	122.7	42.8	74.7	359.0	152.4	216.9
BCD	644.3	279.2	1067.2	1072.6	427.4	1108	5050	2222	773.5
MADC	50.7	20.8	49.7	121.2	42.3	87.0	143.4	141.5	112.7

Fig. 13 further plots the PSNR and runtime curves on the remaining images with r traversed in $\{0, 2, \ldots, 16\}$. In this figure, although the curves on different images are relatively stable along with the variations of r, the performance of MADC with $r > 0$ is always better than that when $r = 0$. Interestingly, instead of increasing runtime, the efficiency of MADC shows a trend of improvement even when the cost function is more nonconvex. We attribute this to the superiority of truncated nuclear norm. That is, when $r > 0$, more intrinsic rank information can be hold along with the increasing iterations, which then leads to a faster convergence. In practice, we suggest $r = 9$ due to its appealing results on both of effectiveness and efficiency.

VI. CONCLUSION

The priors of low-rank and total variation have achieved great success in the field of image restoration. In this paper, we propose a multichannel completion model under these two constraints for estimating missing elements in natural images. By digging into the rank structure of each color channel and each tensor mode, we discover that the pixel matrix has even lower rank than the corresponding tensor mode. Besides, both the total variation and rank properties of different channels share similar distributions. Based on these analyses, a jointly multichannel optimization scheme is proposed by taking both efficacy and efficiency into consideration. Moreover, we introduce the enhanced low-rank and total variation constraints, i.e., truncated nuclear norm and l_p norm, for better performance.

As the introduced constraints are nonconvex and nonsmooth, we develop a Moreau approximated difference-of-convex (MADC) method pursuing the best solution. The distinctive idea of this method is to approximate one of the nonconvex and nonsmooth regularizers by its Moreau envelope. The approximation function can then be minimized by certain proximal gradient algorithms with majorization and extrapolation techniques. We also prove that the sequence generated by this method is bounded and clusters at a stationary point under mild condition. Extensive experiments on color image completion of several samples with various resolutions demonstrate that, compared to state-of-the-art methods, MADC not only achieves better recovery results, but also performs in a more efficient manner along with different mask ratios.

APPENDIX

A. Proof of Proposition 1

Proof. From the definition of Moreau envelope, we have $m_\gamma h(\boldsymbol{x}) \leq \frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{y}\|^2 + h(\boldsymbol{y})$ for all $\boldsymbol{y} \in \Re^n$. Specifically, $\boldsymbol{y} = \boldsymbol{x}$ leads to $m_\gamma h(\boldsymbol{x}) \leq h(\boldsymbol{x})$. Also, $m_\gamma h(\boldsymbol{x}) \geq 0$ holds due to its definition as the minimum of a non-negative function.

B. Proof of Proposition 2

Proof. From the given assumptions, the following inequality holds:

$$\frac{1}{2\gamma_t}\left\|\boldsymbol{x}^t-\boldsymbol{\zeta}^t\right\|^2 + \inf h \leq \frac{1}{2\gamma_t}\left\|\boldsymbol{x}^t-\boldsymbol{\zeta}^t\right\|^2 + h(\boldsymbol{\zeta}^t) = m_{\gamma_t} h(\boldsymbol{x}^t) \leq \frac{1}{2\gamma_t}\left\|\boldsymbol{x}^t-\boldsymbol{x}^*\right\|^2 + h(\boldsymbol{x}^*).$$

Thus, we have $\boldsymbol{\zeta}^t \in \mathrm{dom}h$ for any t and

$$\left\|\boldsymbol{\zeta}^t-\boldsymbol{x}^*\right\| \leq \left\|\boldsymbol{\zeta}^t-\boldsymbol{x}^t\right\| + \|\boldsymbol{x}^t-\boldsymbol{x}^*\| \leq \sqrt{2\gamma_t(h(\boldsymbol{x}^*)-\inf h) + \|\boldsymbol{x}^t-\boldsymbol{x}^*\|^2} + \|\boldsymbol{x}^t-\boldsymbol{x}^*\| \to 0.$$

C. Proof of Proposition 3

Proof. We prove it by contradiction. If $\{\Psi^t\}_{t\in\Gamma}$ is unbounded, we can presume $\lim_{t\in\Gamma}\Psi^t = \infty$ and define $\boldsymbol{\kappa}^*$ with unit norm as

$$\lim_{t\in\Gamma}\frac{1}{\Psi^t}\frac{\lambda D\boldsymbol{X}^t-\boldsymbol{\zeta}^t}{\gamma_{t-1}r_t} = \boldsymbol{\kappa}^*.$$

Then we have

$$\|\boldsymbol{\kappa}^*\| = 1 \;\text{ and }\; D^{\mathrm{T}}\boldsymbol{\kappa}^* = 0. \tag{29}$$

The definition of $\boldsymbol{\kappa}^*$can further lead to

$$\boldsymbol{\kappa}^* = \lim_{t\in\Gamma}\frac{1}{\Psi^t}\frac{\lambda D\boldsymbol{X}^t-\boldsymbol{\zeta}^t}{\gamma_{t-1}r_t} \in \left\{\lim_{t\in\Gamma}\frac{1}{\Psi^t r_t}\boldsymbol{u}^t : \boldsymbol{u}^t \in \partial f_2(\boldsymbol{\zeta}^t)\right\} \subseteq \partial^h f_2(D\boldsymbol{X}^*),$$

where the left inclusion follows from (14) and the right one follows from Proposition 2 as well as definition D2. Taking $\boldsymbol{Y}_1 = 0$ and $\boldsymbol{Y}_2 = \boldsymbol{\kappa}^*$, then $\boldsymbol{Y}_1 + D^T\boldsymbol{Y}_2 = 0$ exists, this contradicts assumption A4. Hence, $\{\Psi^t\}_{t\in\Gamma}$ is bounded.

D. Proof of Proposition 4

Proof. We prove it by contradiction. If $\{r_t\}_{t\in\Gamma}$ is unbounded, we can presume $\lim_{t\in\Gamma} r_t = \infty$ and $\inf_{t\in\Gamma} r_t > 0$. Then the sequences $\{\boldsymbol{\mu}^t/r_t\}_{t\in\Gamma}$ and $\{D^{\mathrm{T}}(\lambda D\boldsymbol{X}^t-\boldsymbol{\zeta}^t)/\gamma_{t-1}r_t\}_{t\in\Gamma}$ would be bounded. Suppose that

$$\lim_{t\in\Gamma}\frac{\boldsymbol{\mu}^t}{r^t} = \boldsymbol{\mu}^* \;\text{ and }\; \lim_{t\in\Gamma}D^{\mathrm{T}}\left(\frac{\lambda D\boldsymbol{X}^t-\boldsymbol{\zeta}^t}{\gamma_{t-1}r_t}\right) = \boldsymbol{\chi}^*, \tag{30}$$

for some $\boldsymbol{\mu}^*$ and $\boldsymbol{\chi}^*$. Notice that

$$1 = \|\boldsymbol{\mu}^*\| + \|\boldsymbol{\chi}^*\|. \tag{31}$$

Furthermore, with the first relation of (22), we can define certain $\boldsymbol{\xi}^t$ satisfying $\|\boldsymbol{\xi}^t\| \leq \varepsilon_{t-1}$, there exist

$$\boldsymbol{\xi}^t = \nabla l(\boldsymbol{X}^t) + \boldsymbol{\mu}^t + \frac{D^T(\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t)}{\gamma_{t-1}}. \quad (32)$$

By dividing r_t from both sides of (32) and following the sequence $t \in \Gamma$, we have

$$0 = \boldsymbol{\mu}^* + \boldsymbol{\chi}^*, \quad (33)$$

Since $\mu^t \in \partial f_1(\boldsymbol{X}^t_{l_t})$ and $\lim_{t\in\Gamma} 1/r_t = 0$, then from (30) and (13) we get

$$\boldsymbol{\mu}^* \in \partial^h f_1(\boldsymbol{X}^*), \quad (34)$$

We proceed to prove $\boldsymbol{\chi}^* \in \partial^h f_2(D\boldsymbol{X}^*)$. Following Proposition 3, $\Psi^t = \|(\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t)/(\gamma_{t-1}r_t)\|$ is bounded, then $\lim_{t\in\Gamma}(\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t)/(\gamma_{t-1}r_t)$ exists. This together with (30) leads to

$$\boldsymbol{\chi}^* = D^{\mathrm{T}}\lim_{t\in\Gamma}(\tfrac{\lambda D\boldsymbol{X}^t - \boldsymbol{\zeta}^t}{\gamma_{t-1}r_t}) \in D^{\mathrm{T}}\{\lim_{t\in\Gamma}\tfrac{1}{r_t}\boldsymbol{u}^t : \boldsymbol{u}^t \in \partial f_2(\boldsymbol{\zeta}^t)\} \subseteq D^{\mathrm{T}}\partial^h f_2(D\boldsymbol{X}^*),$$

where the left inclusion follows from (14) and the right one follows from Proposition 2 as well as definition D2. This together with (31), (33), and (34) contradict assumption A4. Thus, $\{r_t\}_{t\in\Gamma}$ is bounded.

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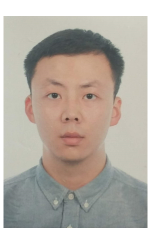

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